Test Plan — Historical VAR Model

Equities Options Portfolio

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1. Overview

This test plan covers the full VAR system described in `requirements.md`. Tests are grouped by layer: data loading, vol surface, portfolio, pricing, scenarios, VAR/ES computation, backtesting, and integration. All tests live in `tests/` and are executed with `pytest`.

Primary testing objectives: - Correctness of mathematical implementations (BS pricing, VAR/ES quantiles, GARCH) - Integrity of the yfinance data loading and vol surface construction pipeline - Correctness of configurable confidence levels for both VAR and ES - Statistical validity of the VAR model under backtesting - Regression safety — changes must not silently break existing behaviour

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2. Test Structure

```
tests/
├── conftest.py                  # Shared fixtures (prices, portfolio, market snapshot)
│
├── unit/
│   ├── test_black_scholes.py         # BS pricer: price, Greeks, edge cases
│   ├── test_loader.py                # yfinance download, fill, cache
│   ├── test_vol_surface.py           # Rolling vol, skew model, surface properties
│   ├── test_portfolio.py             # Trade format, portfolio construction, validation
│   ├── test_scenarios.py             # Log-returns, scenario set, age weights
│   ├── test_revaluation.py           # Full reprice + Greeks approx
│   ├── test_historical_var.py        # Methods A & B — VAR and ES
│   ├── test_filtered_var.py          # Method C — GARCH-filtered VAR and ES
│   ├── test_factor_var.py            # Method D — factor decomposition, VAR and ES
│   └── test_stress_var.py            # Stress window selection and stressed VAR/ES
│
├── integration/
│   ├── test_pipeline_plain_hs.py     # End-to-end: data → plain HS VAR + ES
│   ├── test_pipeline_factor.py       # End-to-end: data → factor VAR + ES
│   └── test_backtest.py              # Rolling backtest + Kupiec/Christoffersen
│
└── fixtures/
    └── sample_data.py                # Deterministic datasets (fixed seed=42)
```

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3. Unit Tests

3.1 Black-Scholes Pricer (`test_black_scholes.py`)

Test ID	Description	Pass Criterion
BS-01	ATM call price (S=K=100, T=1, r=5%, σ=20%)	Within 0.01 of known analytical value (10.45)
BS-02	Put-call parity: C − P = S·e^(−q·T) − K·e^(−r·T)	Difference < 1e-8
BS-03	Deep ITM call → intrinsic value as σ→0	abs(C − max(S−K,0)) < 1e-4
BS-04	Deep OTM call → 0 as σ→0	C < 1e-6
BS-05	Call delta ∈ (0, 1) for all valid inputs	Assertion
BS-06	Put delta ∈ (−1, 0) for all valid inputs	Assertion
BS-07	Gamma > 0 for both calls and puts	Assertion
BS-08	Vega > 0 for both calls and puts	Assertion
BS-09	Theta < 0 for long option (time decay)	Assertion
BS-10	Put price ≥ 0 for all inputs	Assertion
BS-11	Vectorised pricing over strike array returns correct shape	Shape match
BS-12	Zero interest rate (r=0) produces finite, non-negative prices	No exception, price ≥ 0
BS-13	Very short-dated option (T=1/252) — no division by zero	Finite price
BS-14	Very long-dated option (T=10) — finite price	Finite price

3.2 Data Loader (`test_loader.py`)

Test ID	Description	Pass Criterion
DL-01	Download returns a DataFrame with correct tickers as columns	Column set match
DL-02	No NaN in adjusted close after forward-fill + back-fill	isna().sum() == 0
DL-03	All prices > 0 after loading	min > 0
DL-04	Date index is a DatetimeIndex sorted ascending	Sorted check
DL-05	Requesting 5 years returns ≥ 1,200 trading days	len ≥ 1,200
DL-06	Ticker with > 5% missing data emits a warning	warnings.warn called

Test ID	Description	Pass Criterion
DL-07	Cache file written on first download; second call reads cache, not network	File exists; no HTTP call
DL-08	Index tickers (^GSPC, ^NDX, ^RUT, ^VIX) download without error	No exception

3.3 Vol Surface (**`test_vol_surface.py`**)

Test ID	Description	Pass Criterion
VS-01	Rolling 21-day vol output shape = (n_days, n_tickers)	Shape assertion
VS-02	All rolling vol values > 0 after initial warm-up period	min > 0
VS-03	21-day vol ≥ 0 and ≤ 3.0 (300% annualised) for any realistic equity	Range check
VS-04	ATM vol for long-dated tenor uses 126-day window	Equality check
VS-05	ATM vol for short-dated tenor uses 21-day window	Equality check
VS-06	Vol surface shape = (n_days, n_strikes, n_tenors)	Shape assertion
VS-07	No negative vols anywhere in surface	min(σ) > 0
VS-08	Skew at k=0 (ATM) equals σ_ATM	Exact match
VS-09	Skew is negative for k < 0 (OTM put) when α < 0	σ(k<0) > σ_ATM
VS-10	Calendar-spread arbitrage-free: total variance non-decreasing in T	All diffs ≥ 0
VS-11	σ(k, T) for custom (α, β) matches analytical formula	Numerical equality

3.4 Portfolio (**`test_portfolio.py`**)

Test ID	Description	Pass Criterion
PO-01	Portfolio loads from valid CSV without exception	No exception
PO-02	Unknown option_type (“exotic”) raises ValueError	ValueError raised
PO-03	Negative notional raises ValueError	ValueError raised

Test ID	Description	Pass Criterion
PO-04	position_sign not in {+1, −1} raises ValueError	ValueError raised
PO-05	Expiry in the past (relative to trade_date) raises ValueError	ValueError raised
PO-06	Portfolio NPV = sum of (position_sign × price × notional) per trade	abs(diff) < 1e-6
PO-07	Long call has positive delta; long put has negative delta	Sign check
PO-08	Generated synthetic portfolio spans ≥ 10 different underlyings	len(underlyings) ≥ 10
PO-09	Expired option (today > expiry) is valued at 0	V == 0
PO-10	Trade with underlying not in equity universe raises KeyError	KeyError raised

3.5 Scenario Construction (**`test_scenarios.py`**)

Test ID	Description	Pass Criterion
SC-01	Log-returns computed correctly vs manual: log(S_t / S_{t-1})	Max abs error < 1e-12
SC-02	252-day scenario set has exactly 252 rows	len == 252
SC-03	Scenario frame has one column per equity + index	Column count match
SC-04	No NaN or inf in scenario frame	isna + isinf == 0
SC-05	Applying scenario return to today's spot produces stressed spot > 0	All stressed spots > 0
SC-06	Age weights sum to 1.0	abs(sum − 1.0) < 1e-10
SC-07	Age weights monotonically increase toward most recent date	Assertion
SC-08	λ=1.0 produces uniform weights (plain HS)	All weights == 1/252
SC-09	Requesting lookback > available history raises ValueError	ValueError raised

3.6 Revaluation (**`test_revaluation.py`**)

Test ID	Description	Pass Criterion
RV-01	Full reprice P&L = 0 when scenario = flat (no moves)	abs(PnL) < 1e-8

Test ID	Description	Pass Criterion
RV-02	Large spot up shock (+20%): call P&L > 0, put P&L < 0	Sign check
RV-03	Large spot down shock (−20%): put P&L > 0, call P&L < 0	Sign check
RV-04	Greeks approx vs full reprice: relative error < 5% for ±2% spot moves	Relative error
RV-05	P&L vector length = number of scenarios	Shape assertion
RV-06	Portfolio P&L = sum of individual trade P&Ls	abs(diff) < 1e-6
RV-07	Long call + short call same strike → P&L ≈ 0 for any shock	abs(PnL) < 1e-4

3.7 Historical VAR & ES — Plain HS and Age-Weighted (`test_historical_var.py`)

Test ID	Description	Pass Criterion
HV-01	VAR > 0 for non-trivial portfolio (loss convention)	VAR > 0
HV-02	ES ≥ VAR at same confidence level	Inequality
HV-03	VAR_99 > VAR_95	Inequality
HV-04	ES_99 > ES_95	Inequality
HV-05	ES_99 > VAR_99	Inequality
HV-06	10-day VAR = 1-day VAR × √10	abs(diff) < 1e-8
HV-07	10-day ES = 1-day ES × √10	abs(diff) < 1e-8
HV-08	Zero-position portfolio → VAR = 0 and ES = 0	Exact zero
HV-09	VAR_95 uses floor(252 × 0.05) = 12 scenarios at the tail	Index check
HV-10	ES_99 is mean of worst 2 scenarios (floor(252 × 0.01) = 2)	Numerical check
HV-11	Custom confidence level [0.975] returns output keyed “var_97.5_1d”	Key exists
HV-12	Age-weighted VAR ≠ plain VAR when λ < 1	Inequality
HV-13	Age-weighted ES ≠ plain ES when λ < 1	Inequality
HV-14	In high-vol recent period: age-weighted VAR > plain VAR	Directional
HV-15	Output dict contains all expected keys for both measures and both horizons	Key set assertion

3.8 Filtered VAR — GARCH (`test_filtered_var.py`)

Test ID	Description	Pass Criterion
FV-01	GARCH(1,1) parameters: $\alpha + \beta < 1$ (stationarity)	Constraint
FV-02	Conditional vol series is positive throughout	$\min(\sigma_t) > 0$
FV-03	Standardised residuals have approximately unit variance	abs(std − 1) < 0.15
FV-04	Filtered VAR > plain VAR in high-vol regime	Directional
FV-05	Filtered VAR < plain VAR in low-vol regime	Directional
FV-06	Filtered ES ≥ filtered VAR at same confidence	Inequality
FV-07	Output contains VAR and ES keys at all confidence levels	Key set assertion

3.9 Factor VAR (`test_factor_var.py`)

Test ID	Description	Pass Criterion
FA-01	OLS regression $R^2 > 0.3$ for a market-correlated equity	R^2 check
FA-02	Beta to ^GSPC is positive for a long large-cap equity	$\beta > 0$
FA-03	Factor VAR for $\beta=1$, single stock, single index ≈ plain HS VAR	Within 10%
FA-04	Factor ES ≥ factor VAR at same confidence	Inequality
FA-05	Portfolio with $\beta \approx 0$ (market-neutral) → near-zero factor VAR	VAR < threshold
FA-06	Residual idiosyncratic P&L has near-zero correlation with factor P&L	abs(corr) < 0.05
FA-07	Output contains VAR and ES keys at all confidence levels	Key set assertion

3.10 Stress VAR (`test_stress_var.py`)

Test ID	Description	Pass Criterion
SV-01	Stress window start/end dates are within the full 5-year history	Date bounds

Test ID	Description	Pass Criterion
SV-02	Stress window has exactly 252 trading days	len == 252
SV-03	Stressed VAR ≥ plain HS VAR (stress is worst-case)	Inequality
SV-04	Stressed ES ≥ stressed VAR	Inequality
SV-05	Injecting an artificial extreme-loss period → stress window selects it	Exact match
SV-06	Stress window is the same regardless of confidence level input	Stability check

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4. Integration Tests

4.1 End-to-End — Plain HS (**`test_pipeline_plain_hs.py`**)

Test ID	Description	Pass Criterion
E2E-01	Full pipeline runs without exception on 25-name universe	No exception
E2E-02	Output contains VAR and ES at 95% and 99%, 1-day and 10-day	All 8 keys present
E2E-03	All output values are finite positive floats	isfinite and > 0
E2E-04	Doubling all notionals doubles VAR and ES	Ratio ≈ 2.0
E2E-05	Perfectly hedged portfolio (long call + short call, same strike, same notional) → VAR ≈ 0	VAR < 1e-4

4.2 End-to-End — Factor VAR (**`test_pipeline_factor.py`**)

Test ID	Description	Pass Criterion
FA-E2E-01	Factor pipeline runs without exception on 25-name universe	No exception
FA-E2E-02	Factor VAR and plain HS VAR are in the same order of magnitude	Ratio ∈ (0.3, 3.0)
FA-E2E-03	Factor ES ≥ factor VAR in all outputs	Inequality throughout

4.3 Backtesting (**`test_backtest.py`**)

Test ID	Description	Pass Criterion
BT-01	Rolling backtest produces one VAR estimate per business day over test window	len == n_test_days
BT-02	For a correctly calibrated synthetic dataset, exception count $\in [1, 7]$ over 250 days	Basel Green/Amber
BT-03	Kupiec test: p-value > 0.05 for correctly calibrated model (seed=42)	$p > 0.05$
BT-04	Christoffersen test: p-value > 0.05 (no clustering) for correctly calibrated model	$p > 0.05$
BT-05	Artificially inflated VAR (1000× actual) → 0 exceptions → Kupiec $p < 0.05$	$p < 0.05$
BT-06	Traffic light = “Green” when exceptions ≤ 4	String match
BT-07	Traffic light = “Amber” when exceptions $\in [5, 9]$	String match
BT-08	Traffic light = “Red” when exceptions ≥ 10	String match
BT-09	ES backtest: average loss in exception days > ES estimate (conservative check)	Directional

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5. Edge Case & Robustness Tests

Test ID	Description	Pass Criterion
EC-01	Single-trade portfolio (one call) — pipeline runs without error	No exception
EC-02	All options expire on the same date — correct aggregation	No exception
EC-03	Portfolio with mix of calls and puts on same underlying	No exception
EC-04	Very deep OTM option (K = 2×S) priced correctly	Price ≥ 0, finite
EC-05	Very deep ITM option (K = 0.5×S) priced correctly	Price ≥ intrinsic
EC-06	25-name, 100-option portfolio end-to-end completes in < 60s	Timing assertion
EC-07	Confidence levels list with a single value [0.99] works correctly	No exception

Test ID	Description	Pass Criterion
EC-08	Confidence levels list with three values [0.90, 0.95, 0.99] works correctly	All 12 output keys present
EC-09	Missing ticker in yfinance (delisted) triggers a clear error or skip with warning	Exception or warning raised

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6. Test Fixtures

All unit tests use fixed random seed (seed=42) for reproducibility.

conftest.py provides session-scoped fixtures:

```
sample_prices          # 5-year daily price DataFrame, 5 equities, seed=42
sample_index_prices    # 5-year daily prices for ^GSPC, ^NDX, ^RUT, ^VIX
sample_vol_surface     # Vol surface matching sample_prices tickers
sample_portfolio       # 8-trade options book (3 calls, 3 puts, 2 spreads)
today_market           # Market snapshot (spot, vol surface) for revaluation tests
```

Integration test fixtures (session-scoped, generated once):

```
full_prices            # 5-year, 25 equities + indices
full_portfolio         # 50-trade synthetic book across 10 underlyings
```

fixtures/sample_data.py contains hard-coded, deterministic small datasets for tests that must not touch the network.

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7. Test Execution

```
# Run all tests
pytest tests/ -v

# Unit tests only
pytest tests/unit/ -v

# Integration tests only (slower, touches more data)
pytest tests/integration/ -v

# Run with coverage report
pytest tests/ --cov=src --cov-report=html

# Run a specific file
pytest tests/unit/test_black_scholes.py -v

# Run a specific test by ID pattern
```

```
pytest tests/ -k "BS-02 or HV-05" -v

# Skip network-dependent tests
pytest tests/ -m "not network" -v
```

Mark network-dependent tests (yfinance calls) with `@pytest.mark.network` so they can be skipped in offline CI.

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8. Acceptance Criteria (Definition of Done)

Before the base model is considered implementation-complete:

- ☐ All unit tests pass (0 failures, 0 errors)
- ☐ All integration tests pass
- ☐ Code coverage ≥ 80% across `src/`
- ☐ Both VAR and ES outputs present at 95% and 99% for all four methods
- ☐ Configurable confidence level list passes through correctly (EC-08)
- ☐ Backtesting: Kupiec p-value > 0.05 on synthetic dataset (seed=42, 99% CL)
- ☐ Rolling 252-day backtest exception count in Basel Green or Amber zone (≤ 9)
- ☐ End-to-end pipeline (25 names, 50 options) completes in < 60 seconds

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End of test_plan.md